Miramin Bagheri

Senior AI Engineer

Agentic Systems & Enterprise Platforms



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PROFILE

Hands-on AI engineer with **5+ years of experience** spanning custom model development, enterprise architecture and production operations. Principal engineer of an ERP-integrated platform serving **600 organizational users**. Build agentic systems that connect organizational data to real workflows, combining software engineering, applied AI and technical leadership to make fragmented business processes work together.

600	5+	~100
Organizational users	Years of experience	Projects across career

ENTERPRISE AI & PLATFORM ENGINEERING

Tarariss

Jul 2025 - Present

AI Engineering Lead

- **Enterprise platform architecture.** Designed and developed a **modular enterprise application** integrated with an existing **SQL Server ERP**. Unified messaging, employee services, approvals, operational forms and analytics across web, installable Windows and Android clients, for on-site and remote use.
- **Embedded operational agents.** Built a custom **OpenAI API agent harness** inside the enterprise messenger, translating natural-language requests into ERP reporting, data entry and business workflows through tools that respect each user's access permissions.
- **ERP integration through MCP.** Co-developed the ERP MCP integration for authorized AI clients; implemented **caller-identity checks, privacy filters and static write-safety verification** for reporting and ERP data entry.
- **Shared knowledge for coding agents.** Built the team's **Codex / Claude plugin**: onboarding loads ERP knowledge, connection guidance and repository rules; offboarding records changes and findings in shared documentation and the ERP knowledge base, preserving context across developers and sessions.
- **Production maintenance & delivery.** Maintain application, data and deployment layers, resolving integration failures and adapting services to ERP schema changes. Built an overnight job to detect ERP changes and regressions, apply fixes, consolidate team merge requests, merge into main and deploy to production.
- **Hands-on technical leadership.** Lead **two direct reports** while contributing directly to architecture and implementation. Review code, guide technical decisions and coordinate GitLab merge requests and releases.

Further applied AI delivery

Meeting-to-ERP pipeline with approved minutes and action follow-up; unmatched-deposit identification; retention modelling, leave decision support, applicant onboarding, training and machine-stop workflows; industrial vision for a 193-camera estate, including PhoneWatch and a FireWatch pilot; and a live procurement agent for requester clarification, supplier negotiation, follow-up and purchase prioritization.

CUSTOM MODEL DEVELOPMENT

AI Developer · Computer Vision

2023-2024

Confidential client · Germany / Remote · NDA

- Developed object detection for **aerial imagery and autonomous-driving HD maps**, working with large-scale datasets and transformer-based vision architectures in an international engineering team.
- Extended **DINO Swin-Large in Detrex and Detectron2**, implementing a custom detection head and task-specific loss computation. Resolved a core architecture bottleneck that had stalled the project, enabling stable training and inference for the target task.
- Built **COCO data pipelines** and high-resolution image tiling, performed GeoJSON annotation quality checks, and ran large-scale training on **AWS EC2 A100 GPUs**.

TEACHING & MENTORING

AI & Programming Instructor · Rahbord Academy 2024-2025

Designed and taught project-based Python, machine-learning and deep-learning courses; mentored practical student projects.

Teaching Assistant · University of Tehran 2023-2025

Supported programming, machine-learning and deep-learning instruction and student projects.

TECHNICAL CAPABILITIES

AI & agent engineering

Agent harnesses · Tool / Function Calling · MCP · RAG · Structured Outputs · Grounding & Validation · LLM Evaluation · OpenAI / Gemini APIs · Whisper STT

Machine & deep learning

PyTorch · TensorFlow · Keras · scikit-learn · XGBoost · Transformers · CNN / RNN · NLP · Time-Series Forecasting · Explainable AI · Algorithmic Fairness

Computer vision

YOLO · Detectron2 / Detrex · DINO Swin-Large · Object Detection · Tracking · Segmentation · Pose Estimation · Face Recognition · OCR · RTSP / CCTV · Image Tiling

Backend & applications

Python · FastAPI · REST APIs · HTML / CSS / JavaScript · PWA · Service Workers · Web Push · Android (Java) · PyQt6 · Windows Services · Job Scheduling

Data & integration

SQL Server · T-SQL · SQLite · Data Pipelines / ETL · ERP Integration · Schema Drift & Impact Analysis · Data Reconciliation · Analytics & Reporting

Delivery & security

Linux · Docker / Compose · Git · GitLab CI/CD · Code Review · Production Maintenance · JWT / OTP · OAuth · Prometheus · Threat Modelling · Security Auditing

EDUCATION & RESEARCH

MSc · Industrial Engineering · Systems Optimization

University of Tehran · 2021-2024 · **GPA 17.65 / 20**

Thesis: Automated deep-learning framework for **multiple-sclerosis lesion segmentation using FCU-Net**.

LANGUAGES

Persian · Native

English · Advanced (C1)